

Problem 8.12

In Account A , an investment of \$1,000 grows to \$1,700 in four years at an effective annual interest rate of x . In Account B , \$3,000 is invested for five years, at an effective rate of discount $d = x$. What is the ending balance in Account B ?

Answer

For Account A ,

$$1000(1+x)^4 = 1700.$$

Thus,

$$(1+x)^4 = 1.7,$$

so

$$x = (1.7)^{1/4} - 1 = 0.141858.$$

Since Account B has effective rate of discount $d = x$, its accumulation factor is

$$\frac{1}{1-d}.$$

Therefore, if y is the ending balance in Account B ,

$$y = 3000 \left(\frac{1}{1-d} \right)^5.$$

Using $d = x = 0.141858$,

$$y = 3000 \left(\frac{1}{1-0.141858} \right)^5 = 6446.5348.$$

Hence, the ending balance in Account B is

$$\boxed{\$6,446.53}.$$